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Chapter 4

Tips And Techniques



Job Queues And Subsystems

This section deals with the types of job queues and subsystems that are routinely used in Machine Manager.

Job Scheduler

All Day-Start and Day-End Jobs are submitted to the Machine Manager Job Scheduler. This controls their execution, either running them on the next activation of the Scheduler or at the required time. The scheduler may be activated by the MMSTRPGM System Start Up program or by the RUNADSJOB command by **F21** from the Machine Manager main menu. The Scheduler is submitted as a batch job and should be submitted to your controlling system, normally QCTL. The queue to be submitted to is controlled via Job Description MMJOBSCH.

If for whatever reason the scheduler is not running and you wish to start it you may use the command SBMJOBSCH. This command re-starts the job scheduler. It will not re-schedule the Day Start jobs. However, it will process any jobs that are on the job schedule. To examine jobs on the Job Schedule, see the Job Schedules chapter.

You may experience authority problems when the scheduler attempts to submit jobs. This occurs because the Scheduler is run by user profile QPGMR and is attempting to submit jobs (SBMJOB) for User Profiles it does not have authority to. This problem can be overcome by changing the job description MMJOBSCH to have a user that does have authority to use other User Profiles.

To do this enter the command:

```
CHGJOB IPGMMP4/MMJOBSCH USER(aaaaaaaaaaaa)
```

Day End To Day Start

The batch-to-interactive configuration is controlled by the day-start procedure. This can be performed either as part of a system IPL or by running the day-start job itself.

If you wish to use Machine Manager to power down and IPL your machine, then the command SBMPWRDWN is supplied. This calls another command, RUNPWRDWN, to perform the power down, using the job description PWRDWNSYS. The job description directs the submitted job to job queue PWRDWNSYS. If the power down is to be the last job performed, then this job queue must be added as the last job queue entry in the batch subsystem.

To do this, enter the following command:

```
ADDJOBQESBSD(QBATCH)JOBQ(IPGMMP4/PWRDWNSYS)SEQNBR  
(number)
```

Where the number is higher than any of the other job queue entries.

Alternatively, if you do not wish to power down the machine, then you must run the day-start routine as the last day-end job. This can be performed by using the command RUNADSJOB.



Note: If you have created a system library as part of the installation routine then the job queue and job description PWRDWNSYS will be duplicated into the system library. For ease of use add the system library PWRDWNSYS job queue as the job queue entry and change the system library PWRDWNSYS job description to point at this job queue. Details of how to do this are contained in the Software Installation Product Guide.

Day Start To Day End

Machine Manager has been designed to provide a flexible approach to the automatic management of day-time (primarily interactive) processing, and night-time (predominately batch) processing. This is in effect what the auto day-start and auto day-end functions do. The interactive-to-batch switch is controlled by auto day-end. The batch-to-interactive interface is controlled by the auto day-start function, which may involve a power down.

The interactive-to-batch configuration is controlled by running the command `SBMENDDAY` as the last function in the day-start procedure. This submits a job to the Job Scheduler to execute the command `RUNENDDAY`. The submission is controlled by job description `ENDDAY`.

Standard Day Start Jobs

The following are supplied as Standard Day-Start Jobs. See the Standard Jobs section.

Description	Module	Request Data
Start SBS QINTER	MM	STRSBS SBSD(QINTER)
Hold JOBQ PWRDWNSYS	MM	HLDJOBQ JOBQ(PWRDWNSYS)
Change SBS QBATCH to active jobs *NOMAX	MM	CALL PGM(CHGQBATCH)
	MM	PARM (*NOMAX)
Start SBS QBATCH	MM	STRSBS SBSD(QBATCH)
Start SBS QPGMR	MM	STRSBS SBSD(QPGMR)
Start SBS QSYSOPR	MM	STRSBS SBSD(QSYSOPR)
Start SBS QSPL	MM	STRSBS~SBSD(QSPL)
Start SBS QCMN	MM	STRSBS SBSD(QCMN)
Start SBS QSNADS	MM	STRSBS SBSD(QSNADS)
Start printer SYSPRT	MM	STRPRTWTR DEV(SYSPRT)
Start spool distributors	MM	STRSBS SBSD(DSTSPLF)
Initiate ENDDAY	MM	SBMENDDAY
Delete ADS overrides	MM	DLTADSOVR
Start Clean Up	MM	QSYS/STRCLNUP
NW = Network Manager MM = Machine Manager		

Standard Day End Jobs

The following jobs are supplied as Standard Day-End Jobs. See the Standard Jobs section.

Description	Module	Request Data
Change SBS QBATCH to active jobs 1		Call PGM(CHGQBATCH) PARM('1')
Release JOBQ PWRDWN SYS	MM	RLSJOBQ JOBQ(PWRDWN SYS)
Clear output queues	MM	CLRRUNPOUTQ
Clear message queue	MM	CLRRUNPMMSGQ
Clear job queues	MM	CLRRUNOUTQ
End spool file distributors	NW	ENDDSTPLF ENDSBS(*YES) DELAY(600)
File re-organisation	HK	RGZALLPFMS
Compress libraries	HK	COMPALLPGM
Size all libraries	HK	CALL PGM(HK903CLP)
Library/archive saves to save files	HK	CALL PGM(HK900CLP)
Folder saves to save files	HK	CALL PGM(HK970CLP)
Save group saves to save files	HK	CALL PGM(HK975CLP)
Delete unknown libraries	HK	CALL PGM(HK600CLP)
Application Manager day-end	AM	CALL PGM(XA803CLP)
Clear Network Log	NW	CLRNETLOG KEEP(7)
Clear Machine Manager Log	MM	CLRMNGLOG KEEP(7)
Power Down system	MM	SBMPWRDWN
Run day-start procedure	MM	RUNADSJOB
Delete ADE overrides	HKMM	DLTADEOVR
NW = Network Manager MM = Machine Manager HK = Housekeeping AM = Application Manager		

Machine Manager With Subsystem QBASE

On delivery from IBM, the AS/400 is configured to use subsystem QBASE for all batch and interactive jobs. Most users will change the system configuration to use subsystem QINTER for interactive jobs and QBATCH for batch jobs. Machine Manager, as delivered, assumes this to be the case. For smaller users the system may not be re-configured. On machines where QBATCH and QINTER are not being used, Machine Manager needs to be changed.

Power Down Submission

Subsystem QBASE will have MAXJOBS(*NOMAX) specified and be fed by several job queues. Whichever queue is to be used to submit the day-end jobs to batch, the Power Down job should also be submitted to. This would normally be job queue QBATCH. To do this, change job description PWRDWNSYS to submit to this queue. The run priority should be 9. To run successfully, job queue QBATCH should be set to MAXJOBS(1) in its job queue entry of QBASE.

Machine Manager Power Down Parameters

The power down parameters for your processor are held in data area MMPWRDWN in library IPGMMP4. This is a 250 character data area containing the normal and override parameters. The format is:

Description	Start Position	Length
Option (*CNTRLD/*IMMED)	1	7
Delay Time	8	8
Restart (*YES/*NO)	16	4
Power Down Limit 1 (HHMMSS)	29	6
RUNADSJOB (*YES/*NO)	35	4
Override Option	41	7
Override Delay	48	8
Override Restart	56	4
Override Power Down Limit 1	69	6
Override RUNADSJOB	75	4
Last SBMPWRDWN Date (YYMMDD)	81	6
Last SBMPWRDWN Time (HHMMSS)	87	6
Last SBMPWRDWN Date (YYMMDD)	93	6
Last SBMPWRDWN Time (HHMMSS)	99	6
Last SBMPWRDWN Action (1=PWRDWN SYS, 2=RUNADSJOB)	105	1
Power Down Job I.D. (Job Name/User/Job No.)	106	26
Message Queue/Library	132	20
IPL Date (YYMMDD)	161	6
IPL Time (HHMMSS)	167	6
Override IPL Date (YYMMDD)	173	6
Override IPL Time (HHMMSS)	179	6
Power Down Limit 2 (HHMMSS)	185	6
Override Power Down Limit 2 (HHMMSS)	193	6
Power Down Time (HHMMSS)	201	6
Override Power Down Time (HHMMSS)	207	6